Daniil Vlasenko

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Research Interests: Neuroscience, Network Science, Statistical Data Analysis, Probabilistic Modeling

Education

HSE University, Institute for Cognitive Neurosciences
PhD, Doctoral School of Cognitive Science
Moscow, Russia, November 2025 – present

HSE University, Institute for Cognitive Neurosciences
MS, Cognitive Sciences and Technologies
Moscow, Russia, September 2023 – June 2025

Saint Petersburg State University, School of Mathematics and Mechanics, Department of Statistical Modelling
BS, Applied Mathematics and Computer Science
St. Petersburg, Russia, September 2019 – June 2023

Additional Education

Neuromatch Academy, NeuroAI course
Online, 14 - 25 July 2025

Neuromatch Academy, Computational Neuroscience course
Online, 8 - 26 July 2024

Mediterranean School of Complex Networks
Grado, Italy, 30 June - 5 July 2024

Bioinformatics Institute, Professional retraining program "Algorithmic Bioinformatics"
St. Petersburg, Russia, September 2022 – January 2023

Selected Publications

(Q1 Scientific Journal) Krivonosov, M., Nazarenko, T., Ushakov, V., Vlasenko, D., Zakharov, D., Chen, S., Blyus, O., & Zaikin, A. (2025). Analysis of Multidimensional Clinical and Physiological Data with Synolitical Graph Neural Networks. Technologies, 13(1), 13.

Vlasenko D., Saranskaia I., Zakharov D., "From Pairwise to High-Order: Hypergraph Methods and Multivariate Connectivity Metrics for EEG/MEG," 2025 7th CNN, Nizhny Novgorod, Russian Federation, 2025, pp. 142-145.

Additional publications can be found on my Google Scholar page.

Selected Conferences

Volga Neuroscience Meeting 2025, The 7th International Conference “Neurotechnologies and Neurointerfaces”, topic “From Pairwise to High-Order: Hypergraph Methods and Multivariate Connectivity Metrics for EEG/MEG” (poster presentation)
Nizhny Novgorod, Russia, 25 - 29 August 2025

International School and Conference on Network Science: **NetSci 2025**, topic “Ensemble-Based Graph Representation of fMRI Data for Cognitive Brain State Classification” (poster presentation)
Maastricht, the Netherlands, 2-6 June 2025

Baltic Forum 2024: Neuroscience, Artificial Intelligence and Complex Systems, VIII Scientific School “Dynamics of Complex Networks and their Applications”, topic “*Ensemble methods for representation of fMRI, EEG/MEG data in graph form for classification of brain states*” (poster presentation).
Kaliningrad, Russia, 19–21 September 2024

Scholarships and Fellowships

Combined Master's-PhD track at HSE University, the talent program designed for graduate students enrolled on full-tuition scholarships.
September 2023 – present

Academic Development Program (New Scientist category) at HSE University, the talent management program aimed at supporting the professional development of promising teachers and researchers.
January 2024 - May 2025

Honors and Awards

Winner (1st Place), Information Technology & Mathematics Section, 10th All-Russian Youth Scientific Forum “Science of the Future – Science of the Youth”.
September 2025.

Certificate of Excellence “**Best Diploma of the Year 2025**”, HSE University Institute for Cognitive Neuroscience.
June 2025

(Selected) Research experience

Russian Science Foundation Interdisciplinary Grant 24-68-00030 "Next Generation Cognitive Artificial Intelligence"
Research-assistant
June 2024 - present

My tasks: development of representation methods of fMRI, EEG/MEG data in graph form and application of graph neural networks for a classification task of brain states.

Technologies: python, scikit-learn, pytorch geometric, igraph, nilearn, mne, scipy, numpy, pandas; R, dplyr, tidyr, ggplot2.

HSE University, Institute for Cognitive Neurosciences, Strategic Project "Human Brain Resilience", subproject "AI-based listening diagnostic systems"
Research-assistant
September 2023 – December 2024

My tasks: development of neuromorphic dynamic models of speech information processing based on cross-frequency interaction of macroscopic brain rhythms.

Technologies: python, numpy, pandas, matplotlib, scipy, jax, imit_utils, syllabify, brian2, brian2hears.

Teaching Experience

Supervision of course and diploma works of undergraduate students, **HSE University**
September 2024 – present

Seminar instructor (a few classes per course), “Advanced Computational Neuroscience”, “Some Topics of Calculus, Differential Equations and Nonlinear Dynamics”, **HSE University**
September 2024 – present

Languages

Russian (native), English (B2)

Skills

Programming
Python — NumPy, Pandas, Matplotlib, SciPy, scikit-learn, PyTorch, PyTorch Geometric, igraph, nilearn, mne; R — dplyr, tidyr, ggplot2, igraph; C++; MySQL; algorithms and data

structures; Linux-based operating systems; familiar with remote server operation (terminal, Bash), version control systems (git, GitHub), HTML, CSS, JavaScript and Selenium; preparing documents and presentation slides using LaTeX.

Data analysis and machine learning
Classical data analysis; analysis of categorical data; network analysis; machine learning including deep learning; databases (SQL queries). Analysis and preprocessing of fMRI, EEG/MEG data; analysis of neural electrode data (neuropixels probes).

Mathematics
Statistics; probability theory; graph theory; algebra; mathematical analysis; analytic geometry; computational mathematics; optimization methods.